voipred.dev: tests and messages

Net benefit, value of information and plots for binary, survival and competing-risks outcomes

18 October 2026

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Summary

The full test suite of voipred.dev (on top of commit 293f9ee) was run on 18 October 2026 with R version 4.3.3 (2024-02-29): **201 tests with 1,528 checks (expectations), 0 failures, 0 skipped, 0 warnings.** It takes about 10 minutes.

The tests check voipred against independent references:

- **dcurves**: for every model, the net benefit equals `dcurves::dca()` on the same predicted risks at every threshold from 0 to 0.99 (tolerance 10^{-10}); Kaplan-Meier for survival and Aalen-Johansen for competing risks.
- **The models' own predictions**: the risk voipred gets from each model equals the model package's own prediction, or an independent calculation for survival models (tolerance 10^{-6}).
- **cIPCW**: the dcurves cIPCW branch (a saved reference), a direct implementation of the estimator, and the true net benefit in simulations with informative censoring.
- **predtools** (`evpi_val()` , the code of Sadatsafavi et al. 2023): EVPI and P(useful) with the same Bayesian bootstrap draws (tolerance 10^{-12}).
- **The EVSI paper's code** (Sadatsafavi et al. 2025, resplab/papercode and resplab/evsiexval): the Bayesian bootstrap EVSI and the conjugate `EVSI()` , with the same random numbers.

- **bayesDCA**: the posterior draws of the net benefit equal bayesDCA's with the same seed.
- **The usual net benefit**: with the default prior, `nb_improper()` (Beta(0, 0), the data only), the net benefit equals TP/n - FP/n z/(1-z) exactly, and its posterior draws equal the Bayesian bootstrap in distribution (means, spread, correlation between models, EVPI, P(best) and EVSI).
- **Hand calculations** and direct loops for the core estimator; the compiled Kaplan-Meier code equals the R implementation.

Part 1 summarizes the tests by area and lists the models checked, with every test listed at the end of the document. Part 2 lists every error and warning message voipred can give.

Part 1: tests

Tests by area

Area	File	What is checked	Reference	Tests	Checks
Net benefit, binary	test-nb.R	Hand calculation; treated if risk >= threshold, risk 0 never treated; thresholds kept in order; vectorized estimator equals a direct loop; outcome as column, vector, logical or factor; integer weights equal duplicated rows; risks given directly; the result object; data first (old argument order refused); input errors; print.	Hand calculation; direct loop	13	45
Predicted risks given directly	test-risk-input.R	predicted_risk (vector or column) gives the same results as the model for binary, survival and competing risks, KM and cIPCW, other priors, and the value of information; labels; risks predicted once and reused by every later step; input errors.	The same model's results	4	27
Agreement with dcurves	test-dcurves.R	Logistic regression, ranger (with tied risks), randomForest and bayespred: model, treat all and treat none at every threshold.	dcurves::dca()	4	17
Binary models	test-models.R	Every supported binary model class gives the same net benefit as dcurves; log-odds, raw scores, decision values, discriminant scores and predicted classes are rejected; matrix models need numeric feature columns.	dcurves::dca()	26	114
Probabilities, not scores or classes	test-probabilities.R	Only 0s and 1s are rejected as classes; exact 0s and 1s among probabilities are accepted; values outside [0, 1] are rejected as scores.	Expected errors	8	16
Risks from a model (internal)	test-predict_risk.R	The risk equals the package's prediction (glm, ranger, randomForest, bayespred); multi-class forests, vectors and objects without predict() are rejected.	Package predictions	12	15
How risks are obtained	test-prediction-routes.R	predict_function takes precedence; it gets horizon and cause and is checked like any model; output shapes (vector, class columns, survival curve); order of the standard routes; a new model works with predict_function; the prediction code is internal (not exported).	Expected values and errors	8	37
Missing covariates	test-covariates.R	Missing covariates are named (glm, formula y ~ ., transformed covariates, randomForest, bayespred); ranger's own error is shown; extra columns are fine.	Expected errors	8	14

Area	File	What is checked	Reference	Tests	Checks
Survival outcome and arguments	test-survival-outcome.R	Outcome as formula, Surv object or column; horizon, cause, censoring and cipcw arguments are checked; only right censoring; missing or negative times.	Expected values and errors	11	43
Kaplan-Meier / Aalen-Johansen	test-survival-km.R	Net benefit equals dcurves at every threshold, including NA thresholds; internal KM and Aalen-Johansen equal survival::survfit (with weights); weights equal duplicated rows; KM biased under informative censoring.	dcurves::dca(); survival::survfit; simulation truth	10	100
Compiled Kaplan-Meier code	test-km-compiled.R	The compiled KM / Aalen-Johansen net benefit equals the R implementation (tolerance 10^-14), with random, zero and integer weights, ties and early horizons.	R implementation	1	162
Survival models	test-survival-models.R	Each survival model class: risk equals an independent calculation and the net benefit equals dcurves; models without probabilities are refused (also a model whose formula is in a variable); binary models with a survival outcome warn; horizon beyond follow-up warns; risk grows with the horizon.	Independent risk; dcurves::dca()	19	215
Competing risks	test-survival-competing.R	CSC, FGR, crr, competing-risks random forest and multi-state coxph: risk equals the package's cumulative incidence and the net benefit equals dcurves (Aalen-Johansen); causes by label or position; FGR fitted for another cause is refused.	Package CIF; dcurves::dca()	10	97
Conditional IPCW	test-cipcw.R	Equal to the dcurves cIPCW branch; equal to a direct implementation; a model gives the same as its weights by hand; close to the truth under informative censoring (also competing risks); censoring model checks; truncation and positivity; ties; case weights; every censoring model class; censoring predict_function.	dcurves cIPCW branch; direct code; simulation truth	20	129
The prior of nb()	test-bayes-nb.R	Posterior means equal the Beta model by hand for the uniform, Jeffreys and custom priors; the default improper prior (the data only) equals the usual net benefit exactly (also with case weights); counts of 0 handled as the usual estimate; prior constructors and printing; plots refuse different priors; weights only with the improper prior; survival only with the improper prior; input errors.	Hand calculation	4	46

Area	File	What is checked	Reference	Tests	Checks
Value of information	test-voi.R	Draws with weights 1 reproduce nb() for every outcome and estimator; EVPI and P(useful) equal predtools::evpi_val(); Bayesian bootstrap EVSI equals the paper's algorithm; P(useful), P(best), relative EVPI by their definitions; survival without censoring gives the binary results; case weights; NA thresholds; unknown arguments (no `method`); p_useful() takes one model; input errors.	predtools; EVSI paper code; definitions	11	87
cIPCW refit in each draw	test-voi-refit.R	Every known censoring model (coxph, survreg, cph, psm, flexsurvreg, prodlim) is refitted with the draw's weights (equal to duplicated rows); other models need refit_function (equal to duplicated rows); shared refits; EVPI/EVSI with refits, also competing risks; given weights refused.	Duplicated rows; direct refits	7	49
Value of information: posterior draws	test-bayes.R	One model: posterior draws and P(useful) equal bayesDCA; EVSI equals the conjugate EVSI() of the paper; several models are drawn jointly, each keeping its own posterior; multinomial draws; different priors not compared; with the improper prior the posterior draws equal the Bayesian bootstrap in distribution.	bayesDCA; EVSI paper code; Bayesian bootstrap	6	77
Result objects: print(), summary(), as.data.frame()	test-methods.R	Each result is a list (not attributes) with its parts; print() gives a header and the first rows; summary() gives the best strategy, sensitivity and specificity (share treated for survival), EVPI and the best strategy now, EVSI as % of EVPI, P(useful) bands and the most likely best strategy, checked by hand; chosen thresholds; as.data.frame() is the table.	Hand calculation; the results	5	113
plot() of nb() and delta_NB	test-plot.R	delta_NB = NB_model - max(NB_all, NB_none); labels; plot data equal the results; several results together; colours; incompatible results refused.	nb() results	6	45
VoI plots, interactive plots, DCA with EVPI/EVSI	test-plot-voi.R	plot() of evpi() (EVPI, relative EVPI), evsi() (chosen thresholds), p_useful() (one model per result, several results on one plot), p_best(): curves equal the results; plot() of nb() with EVPI/EVSI below (shared axis, a legend per panel); interactive = TRUE gives the same curves; shared look and legend placement; axis names; plotly-safe labels.	evpi(), evsi(), p_useful(), p_best() results	8	80
Total				**201**	**1528**

Models checked

Each model is fitted on development data and evaluated on validation data (GUSTO-I for binary outcomes, Rotterdam for survival and competing risks). “Standard route” means voipred uses the model’s own `predict()` or `riskRegression::predictRisk()`; “voipred method” means voipred has its own (internal) prediction code for the model because the standard route fails or gives a wrong answer. This is only used when the model is given instead of the predicted risks.

Outcome	Models	Route	Checked against
Binary	glm (logit, probit, cloglog), logistf, mgcv::gam, earth, rpart, partykit::ctree, randomForest, ranger (probability = TRUE), gbm, kknn, naivebayes, xgboost()	Standard route	dcurves::dca() at thresholds 0 to 0.99
Binary	e1071::svm (probability = TRUE), e1071::naiveBayes, nnet, MASS::lda, MASS::qda, glmnet, cv.glmnet, xgb.train, lightgbm, catboost	voipred method	dcurves::dca() at thresholds 0 to 0.99
Binary	bayespred bpmfit and bpm (posterior mean of risk)	Standard route	dcurves::dca(); bayespred’s own posterior mean
Survival	coxph with x = TRUE (Efron, Breslow, case weights, ties, strata, splines), rms::cph, rms::psm, survreg, flexsurvreg, flexsurvspline, ranger, randomForestSRC, prodlim	Standard route	Independent risk by the horizon; dcurves::dca() (KM)
Survival	coxph without x = TRUE, stratified survreg, aorsf, partykit::ctree, rstpm2::stpm2, mgcv::gam (cox.ph)	voipred method	Independent risk (e.g. survival::basehaz, survreg quantiles); dcurves::dca() (KM)
Competing risks	riskRegression::CSC, riskRegression::FGR, competing-risks randomForestSRC	Standard route	Package cumulative incidence; dcurves::dca() (Aalen-Johansen)
Competing risks	cmprsk::crr, survival’s multi-state coxph	voipred method	Package cumulative incidence; CSC; dcurves::dca() (Aalen-Johansen)
Censoring model (cIPCW)	coxph, survreg (also stratified), rms::psm, flexsurvreg, flexsurvspline, ranger, randomForestSRC, aorsf, partykit::ctree, rstpm2::stpm2, mgcv::gam, prodlim (Kaplan-Meier); any other model with predict_function	Standard route or voipred method	The model’s own censoring survival; same net benefit as its weights given by hand

bayespred.dev Bayesian survival models (Cox and parametric) are also supported, as models and as censoring models, but bayespred.dev is a private package, so they are not part of the test suite.

The test suite also checks that models which only give scores, relative risks, times or classes are refused with a clear message (e.g. log-odds from `glm`, `glmnet` and `gbm`; XGBoost `binary:logitraw`; LightGBM raw scores; CatBoost raw values and classes; SVM decision values; LDA scores; `ranger` and `svm` without `probability = TRUE`; `xgboost` survival objectives, `gbm` and `glmnet` Cox models, `rpart` survival trees).

Part 2: error and warning messages

voipred has 127 error messages and 13 warnings. They are listed below by what they are about, as written in the code. Text in braces, such as `{n}` or `{class}`, is filled in with the actual values (e.g. the number of rows or the class of the model).

Arguments of nb() (12)

Type	Message
Error	`data` must be a data frame. The first argument of nb() is `data`: nb(data, predicted_risk, outcome, ...).
Error	`data` must be a data frame.
Error	Give either `predicted_risk` or `model`, not both.
Error	`cipcw` must be made with cipcw_control().
Error	`predict_function` predicts from `model`; it is not used with `predicted_risk`.
Error	Give the predicted risks with `predicted_risk` (or a fitted `model`).
Error	`predict_function` must be a function(model, newdata, horizon, cause).
Error	`thresholds` must be a non-empty numeric vector.
Error	`thresholds` must lie in [0, 1).
Error	`label` must be a single non-empty string.
Error	`label` cannot be "{label}": "Treat All" and "Treat None" are the default strategies.
Error	`weights` must be {n} non-negative numbers with a positive sum.

Outcome (11)

Type	Message
Error	An `outcome` formula must be one-sided, e.g. ~ Surv(time, status).
Error	Could not evaluate the `outcome` formula in `data`: {original error}
Error	Column '{column}' is not in `data`.
Error	`outcome` has length {m} but `data` has {n} rows.
Error	`outcome` has missing values.
Error	A factor outcome must have exactly two levels.
Error	`outcome` must be binary (0/1, logical, or a two-level factor) or a survival outcome made with Surv().
Error	Only right-censored survival outcomes are supported: Surv(time, status). This outcome is of type '{type}'.
Error	`outcome` has {m} rows but `data` has {n} rows.
Error	The survival outcome has missing times or statuses. (For competing risks, give the status as a factor whose first level is censoring, e.g. Surv(time, factor(status, 0:2)); Surv() turns other numeric codes into NA.)
Error	The survival outcome has negative times.

Survival arguments (11)

Type	Message
Error	`horizon` is only used with a survival outcome made with Surv().

Type	Message
Error	`cause` is only used with a competing-risks outcome.
Error	`censoring` and `cipcw` are only used with a survival outcome.
Error	A survival outcome needs `horizon` : the time at which the risk is predicted (in the same units as the survival times).
Error	`horizon` must be a single positive number.
Error	`horizon` ({horizon}) is beyond the follow-up of every patient in `data` (longest follow-up: {longest follow-up}).
Error	`cause` is only used with a competing-risks outcome, i.e. Surv(time, status) with a factor status.
Error	`cause` must be one of the event types of the outcome: {event types}.
Error	`cipcw` settings were given but `censoring = "km"` . Use `censoring = "cipcw"` to apply them, or remove `cipcw` .
Error	`censoring = "cipcw"` needs a censoring model or weights: cipcw = cipcw_control(model = ...) or cipcw_control(weight_event = ..., weight_horizon = ...).
Warning	No patient treated at thresholds {thresholds} is followed up to the horizon ({horizon}), so the risk among them cannot be estimated: NB_model is NA there.

Predicted risks given directly (6)

Type	Message
Error	`predicted_risk` names a column (“{column}”) that is not in `data` .
Error	`predicted_risk` must be numeric: one predicted probability per row of `data` , or the name of the column that holds them.
Error	`predicted_risk` has {m} values but `data` has {n} rows.
Error	{k} of the risks in `predicted_risk` are NA.
Error	`predicted_risk` has values outside [0, 1] (from {min} to {max}), so these are not probabilities (e.g. log-odds must be turned into probabilities first).
Error	`predicted_risk` has only 0s and 1s. These look like predicted classes, not probabilities.

Model and covariates (9)

Type	Message
Error	Could not get predicted risks from this {class} model ({reason}). If a covariate is missing from the data, add it. Otherwise give `predict_function = function(model, newdata, horizon, cause)` to nb(), returning the risk of the event by the horizon.
Error	`model` must be a fitted model. To use predicted risks, give them with `predicted_risk =` instead.
Error	`data` must be a data frame.
Error	The validation data is missing {k} covariates that the {model or censoring model} was developed with: {covariates}.
Error	This {class} model was fitted to a survival outcome: use it with a survival outcome made with Surv() and give `horizon` .
Warning	The model ({class}) was fitted to a binary outcome, not a survival outcome. Its predicted probabilities are taken as the risk of the event by the horizon ({horizon}), which can be misleading if it was not built for this horizon and outcome.
Error	The model ({class}) is a competing-risks model, but the outcome has a single event type. Give a competing-risks outcome: Surv(time, status) with a factor status.

Type	Message
Warning	The model ({class}) was fitted to a standard survival outcome: its risk is used as the risk of '{cause}', but it is not the cumulative incidence of that cause when there are competing events, and can overstate the risk. Use a competing-risks model.
Warning	`horizon` ({horizon}) is beyond the last time in the {class} model's own data ({last time}); its predictions there are extrapolated or carry the last estimate forward.

Predicted risks (8)

Type	Message
Error	A survival curve was returned, but there is no `horizon`.
Error	The model predicts {k} outcome classes; the outcome must be binary.
Error	The model returned classes, not probabilities. Make it return probabilities (or give `predict_function`).
Error	Predicted risks must be numeric.
Error	The model returned {m} predictions for {n} rows. Check for missing values in the covariates.
Error	{k} predicted risks are NA. Check for missing values in the covariates.
Error	The model returned values outside [0, 1] (from {min} to {max}), so these are not probabilities. They may be scores from the model (e.g. log-odds, relative risks or predicted times) that must be turned into probabilities first.
Error	The model returned only 0s and 1s. These look like predicted classes (labels), not probabilities. Make the model return probabilities (e.g. fit ranger or svm with `probability = TRUE`).

Specific models (20)

Model	Type	Message
svm	Error	This svm was fitted without `probability = TRUE`, so it only predicts classes. Refit it with `probability = TRUE` to get predicted risks.
glmnet	Error	This glmnet model has {k} values of lambda. Fit it with a single lambda, or use cv.glmnet().
catboost	Error	The catboost package is needed to predict from this model.
glmnet, xgboost, lightgbm, catboost	Error	Could not read the feature names from this {class} model. Train it on a matrix with column names.
glmnet, xgboost, lightgbm, catboost	Error	This {class} model was trained on a numeric matrix, but these columns of the validation data are not numeric: {columns}. Code them the same way as in the development data (e.g. as dummy columns).
gam	Error	This gam model (Cox PH family) is a survival model: use it with a survival outcome made with Surv() and give `horizon`.
bayespred.dev	Error	This bpmfit model is a survival model: use it with a survival outcome made with Surv() and give `horizon`.
bayespred.dev	Error	This bpm object has no baseline hazard, so it gives no risks. Package the fit at the horizon: posterior(fit, time = {horizon}).
bayespred.dev	Error	This bpm object is a survival model packaged at time = {time}: use it with a survival outcome made with Surv() and horizon = {time}.

Model	Type	Message
bayespred.dev	Error	This bpm object gives risks at time = {time}, but `horizon` is {horizon}. Use horizon = {time}, or package the fit at the horizon: posterior(fit, time = {horizon}).
bayespred.dev	Error	The bayespred.dev package is needed for this survival model.
crr	Error	This crr model is a competing-risks model: use it with a competing-risks outcome, Surv(time, status) with a factor status, and give `horizon`.
crr	Error	This crr model was fitted for failcode {failcode}, but the net benefit is for cause '{label}'.
multi-state coxph	Error	This coxph model is a multi-state (competing-risks) model: use it with a competing-risks outcome, Surv(time, status) with a factor status, and give `horizon`.
multi-state coxph	Error	Could not compute the cumulative incidences from this multi-state coxph model: {original error}. Fit it with `model = TRUE` (or keep its data available).
multi-state coxph	Error	Could not match the cause '{cause}' to the states of the multi-state coxph model ({states}).
FGR	Error	This FGR model was fitted for cause '{fitted cause}', but the net benefit is for cause '{cause}'. Fit FGR with cause = '{cause}'.
coxph	Error	Could not compute survival from this coxph model: {original error}. Fit it with `x = TRUE` (or `model = TRUE`).
survreg	Error	Some strata of the validation data are not among the survreg model's strata.
rstpm2	Error	rstpm2 must be attached to predict from this {class} model: run library(rstpm2) first.

The prior (4)

Type	Message
Error	`{argument}` must be two non-negative numbers: the shapes (a, b) of a Beta prior.
Error	`prior` must be nb_improper(), nb_uniform(), nb_jeffreys() or nb_beta().
Error	For survival and competing-risks outcomes the prior is nb_improper() (the data only); other priors are for binary outcomes.
Error	Case weights can only be used with the improper prior, nb_improper().

cIPCW: cipcw_control() (11)

Type	Message
Error	There is no `refit` setting: the value-of-information functions always refit the censoring model in each draw. `refit_function` is an optional function(data, weights) returning the refitted censoring model.
Error	`refit_function` must be a function(data, weights) returning the refitted censoring model.
Error	`refit_function` needs the censoring `model` it refits; given weights cannot be refitted.
Error	Give either a censoring `model` or the weights, not both.
Error	`predict_function` must be a function(model, newdata, times).
Error	`predict_function` needs the censoring `model` it predicts from.
Error	`weight_event` and `weight_horizon` must be given together.
Error	`{argument}` must be positive, finite numbers without missing values.
Error	`weight_event` and `weight_horizon` must have the same length.
Error	`model` must be a fitted censoring model. For weights, use `weight_event` and `weight_horizon`.
Error	`min_prob` must be a single number strictly between 0 and 1.

cIPCW: censoring model (9)

Type	Message
Warning	Could not verify that the censoring model was fitted to the censoring times (its outcome could not be read from its formula). It must be fitted with the censored patients as events, e.g. Surv(time, status == 0).
Error	The censoring model appears to have been fitted to the event, not to censoring. Refit it with only censoring as the event, e.g. Surv(time, status == 0).
Error	The censoring model appears to have been fitted to the cause of interest ({cause}), not to censoring. Refit it with only censoring as the event, e.g. Surv(time, status == 0).
Error	The censoring model appears to have been fitted to all events together, not to censoring. Refit it with only censoring as the event, e.g. Surv(time, status == 0).
Error	The censoring model appears to have been fitted to a competing cause ({other cause}), not to censoring. Refit it with only censoring as the event, e.g. Surv(time, status == 0).
Warning	Could not verify that the censoring model was fitted to the censoring times of `data` : its outcome does not match. It must be fitted with the censored patients as events, e.g. Surv(time, status == 0).
Warning	The censoring model was fitted to {n fitted} patients, but `data` has {n}. It should be fitted to the censoring times of `data` (the validation sample).
Error	Could not get survival probabilities from this {class} model ({original error}). As a cIPCW censoring model, give it with `predict_function` in cipcw_control(), or give the censoring weights instead.
Error	Unexpected output from riskRegression::predictRisk for this {class} model.

cIPCW: censoring weights (10)

Type	Message
Error	The cIPCW weights have length {m} but `data` has {n} rows.
Warning	{k} cIPCW weights were truncated at 1 / min_prob = {1/min_prob}.
Warning	{k} cIPCW weights are above 100 (censoring probability below 0.01), a sign of near-violations of positivity. Consider truncation with `min_prob` .
Error	The censoring model returned invalid censoring probabilities (missing, or outside [0, 1]).
Warning	{k} censoring probabilities were truncated at min_prob = {min_prob}. This signals near-violations of positivity.
Error	{k} censoring probabilities are exactly 0, which gives an infinite weight (positivity is violated). Set `min_prob` to truncate them.
Warning	{k} censoring probabilities are below 0.01, a sign of near-violations of positivity. Consider truncation with `min_prob` .
Error	The censoring `predict_function` returned a curve of the wrong size: `surv` must be {n} rows by length(time) columns.
Error	The censoring `predict_function` must return one probability per row (or a curve list(time, surv)).
Error	The censoring `predict_function` returned {m} values for {n} rows; it must return one probability of remaining uncensored per row.

Value of information (5)

Type	Message
Error	p_useful() takes one result of nb(): the probability that a model is useful does not depend on the other models. Compute it for each model and draw them together with plot(pu_1, pu_2); use p_best() to compare the models with each other.
Warning	The net benefit could not be estimated in some draws at thresholds {thresholds} (no treated patient followed up to the horizon); the value of information is NA there.
Error	`future_n` must be one or more positive whole numbers: the sizes of the future validation sample.
Warning	The net benefit could not be estimated in some draws at thresholds {thresholds} (no treated patient followed up to the horizon); EVSI is NA there.
Error	`n_sim` must be a whole number of at least 2.

cIPCW: refit in each draw (4)

Type	Message
Error	The cIPCW censoring weights were given directly (cipcw_control(weight_event =, weight_horizon =)), so they cannot be re-estimated in each draw, and keeping them fixed would understate the uncertainty. For the value of information, give the censoring model instead: cipcw_control(model =).
Error	The censoring model ({class}) must be refitted in each draw with the draw's weights (Bayesian bootstrap), and voipred does not know how to fit a {class} model with case weights. Give `refit_function = function(data, weights)` in cipcw_control(), returning the censoring model fitted with these case weights.
Error	Could not refit the censoring model in a draw ({original error}). Give `refit_function = function(data, weights)` in cipcw_control(), returning the censoring model fitted with these case weights.
Error	the model keeps no call to refit it with

print() and summary() (2)

Type	Message
Error	`thresholds` must be one or more of the thresholds of the result.
Error	Threshold {threshold} is not among the thresholds of the result.

Plots (18)

Type	Message
Error	`evpi` must be a result of evpi().
Error	`evsi` must be a result of evsi().
Error	The patchwork package is needed to stack the value-of-information panel under the decision curve; install it, or use `interactive = TRUE`.
Error	`interactive` must be TRUE or FALSE.
Error	The plotly package is needed for `interactive = TRUE`.
Error	Unknown argument `{name}`. The arguments of {function}() are {its arguments}; other arguments must be results of nb().
Error	Only results of nb() can be plotted together. / {function}() needs results of nb().
Error	"Treat All" and "Treat None" cannot be used as labels.

Type	Message
Error	The results have the same label (“{label}”); give each a different one with `labels` or nb(label =).
Error	“{label 1}” and “{label 2}” cannot be used together: different {what differs}. Compute them with nb() on the same data, outcome, horizon, cause, censoring, prior and thresholds.
Error	Two results have the same label (“{label}”); give each a different one with `labels` .
Error	Unknown argument to {function}(). Its arguments are {its arguments}.
Error	`labels` must be one string / {n} strings, one per result.
Error	`thresholds` must be one or more of the thresholds of the result.
Error	Threshold {threshold} is not among the thresholds of the result.
Error	Unknown argument `{name}` . The arguments of {function}() are {its arguments}; other arguments must be results of {function}().
Error	{function}() needs results of {function}().
Error	Two results have the same label (“{label}”); give each a different one with `labels` .

Appendix: every test

File	Test	Checks	Result
test-bayes-nb.R	with another prior, the net benefit is the posterior mean of the Beta model	14	pass
	the improper prior (default) gives exactly the frequentist net benefit	7	pass
	the improper prior handles empty groups (counts of 0) as the frequentist estimate	8	pass
	priors are checked	17	pass
test-bayes.R	one model: posterior draws equal bayesDCA's	5	pass
	EVSI equals the conjugate algorithm of Sadatsafavi et al. (2025)	8	pass
	several models are drawn jointly, each keeping its own posterior	44	pass
	multinomial draws by conditional binomials	3	pass
	results with different priors are not compared; `method` is gone	6	pass
	with the improper prior the posterior draws equal the Bayesian bootstrap in distribution	11	pass
test-cipcw.R	same as the dcurves cIPCW branch: Cox censoring model	4	pass
	same as the dcurves cIPCW branch: given weights	2	pass
	same as the dcurves cIPCW branch: competing risks, Cox censoring model	3	pass
	the vectorized estimator equals a direct implementation (ties, weights, causes)	12	pass
	a censoring model gives the same result as its weights given by hand	4	pass
	with Kaplan-Meier censoring weights, the risk by the horizon is 1 - KM	6	pass
	the censoring model is checked against the outcome	15	pass
	the censoring model's covariates must be in the data	1	pass
	small censoring probabilities: warning, optional truncation, error at 0	10	pass
	ties: left uses the censoring survival just before each patient's time	3	pass
	every censoring model class runs and agrees with its own predictions	28	pass
	cIPCW is close to the truth under informative censoring; KM is not	3	pass
	cIPCW is close to the truth with competing risks and informative censoring	2	pass
	cIPCW with case weights equals duplicated rows	2	pass
	a censoring model with case weights equals duplicated rows and dropped rows	7	pass
	print shows the cIPCW estimator	1	pass
	censoring predict_function: same as the model route, per row or as a curve	6	pass
	censoring predict_function makes any censoring model usable	3	pass
	censoring predict_function: input and output checks	8	pass
	more censoring model classes: psm, flexsurvspline, stratified survreg	9	pass

File	Test	Checks	Result
test-covariates.R	glm: missing covariates are named	2	pass
	one missing covariate uses the singular	1	pass
	transformed covariates are reported by their column name	1	pass
	a model fitted with y ~ . knows all its covariates	1	pass
	ranger: covariates cannot be read generically, so ranger’s own error is shown	2	pass
	randomForest (formula and x/y interface): missing covariates are named	2	pass
	bayespred (bpmfit and bpm): missing covariates are named	3	pass
	extra columns and a missing outcome column are fine for prediction	2	pass
test-dcurves.R	logistic regression (glm) agrees with dcurves	4	pass
	random forest (ranger) agrees with dcurves, including tied risks	4	pass
	random forest (randomForest) agrees with dcurves	5	pass
	bayespred (posterior mean of risk) agrees with dcurves	4	pass
test-km-compiled.R	compiled KM / Aalen-Johansen net benefit equals the R implementation	162	pass
test-methods.R	nb() returns a list with its parts, not attributes	14	pass
	print() and summary() of nb() are short and say what the result is	21	pass
	the value-of-information results are lists with print(), summary() and as.data.frame()	70	pass
	ranges of thresholds are written compactly	4	pass
	summary() has the same columns for every outcome; without censoring Se and Sp equal the binary ones	4	pass

File	Test	Checks	Result
test-models.R	glm with probit and cloglog links agrees with dcurves	8	pass
	Firth logistic regression (logistf) agrees with dcurves	4	pass
	generalized additive model (mgcv::gam) agrees with dcurves	4	pass
	MARS (earth) with a binomial glm agrees with dcurves	4	pass
	classification tree (rpart) agrees with dcurves	4	pass
	conditional inference tree (partykit::ctree) agrees with dcurves	4	pass
	gradient boosting (gbm) agrees with dcurves	4	pass
	k-nearest neighbours (kknn) agrees with dcurves	4	pass
	naive Bayes (naivebayes package) agrees with dcurves	4	pass
	support vector machine (e1071::svm, probability = TRUE) agrees with dcurves	4	pass
	naive Bayes (e1071::naiveBayes) agrees with dcurves	4	pass
	neural network (nnet) agrees with dcurves, factor and 0/1 outcome	8	pass
	linear and quadratic discriminant analysis (MASS) agree with dcurves	8	pass
	svm and naiveBayes ignore missing values in columns the model does not use	3	pass
	glmnet (one lambda) and cv.glmnet agree with dcurves	8	pass
	LightGBM (objective binary) agrees with dcurves	4	pass
	XGBoost (binary:logistic) agrees with dcurves, both interfaces	13	pass
	CatBoost (Logloss) agrees with dcurves, numeric and categorical features	8	pass
	svm without probability = TRUE, or its decision values, are rejected	2	pass
	log-odds from glmnet, gbm and probit glm are rejected as scores	2	pass
	XGBoost raw log-odds (binary:logitraw) are rejected	1	pass
	glmnet with a whole lambda path is rejected	1	pass
	LightGBM raw scores and CatBoost raw values and classes are rejected	3	pass
	LDA discriminant scores are rejected	1	pass
	predicted classes from rpart and naiveBayes are rejected	2	pass
	matrix models: missing or non-numeric feature columns give clear errors	2	pass

File	Test	Checks	Result
test-nb.R	nb() matches a hand calculation	5	pass
	nb() returns one row per threshold with the expected columns	6	pass
	at threshold 0 the model and treat-all both equal the prevalence	2	pass
	a risk exactly at the threshold is treated (p >= z, as dcurves)	1	pass
	a risk of exactly 0 is never treated, even at threshold 0 (as dcurves)	2	pass
	unsorted thresholds are kept in the order given	2	pass
	the fast core agrees with a direct loop on random data	1	pass
	outcome can be a column name, a vector, logical, or a factor	4	pass
	weights: all ones equal no weights, integer weights equal duplicated rows	3	pass
	bad inputs give clear errors	9	pass
	the predicted risks can be given directly, without the model	4	pass
	print shows a header and the table	2	pass
	the first argument is data; the old order gives a clear error	4	pass
test-plot-voi.R	plot() of evpi(): EVPI or relative EVPI against the threshold	10	pass
	plot() of evsi(): against the future sample size	10	pass
	plot() of p_useful() and p_best(): probabilities, in the colours of the decision curve	17	pass
	plot() of nb() with an EVPI / EVSI panel below	8	pass
	interactive = TRUE gives plotly widgets with the same curves	14	pass
	static and interactive plots share one look, with the legend on the right	12	pass
	the interactive delta-NB axis title uses codes plotly can show	3	pass
	the value-of-information axis is named after what it shows	6	pass
test-plot.R	delta_NB is NB_model - max(NB_all, NB_none)	4	pass
	label: default from the model’s name, given, or reserved	7	pass
	decision curve: model, Treat All and Treat None	8	pass
	delta plot: delta_NB of each model, line at 0	3	pass
	several results on one plot, for both types	12	pass
	results that cannot be drawn together are refused	11	pass

File	Test	Checks	Result
test-predict_risk.R	glm: predict_risk() equals predict(type = ‘response’)	1	pass
	a glm whose predictions are not probabilities is rejected	1	pass
	ranger: the risk is the probability of class 1	1	pass
	ranger: numeric 0/1 outcome (unnamed columns) gives the second column	1	pass
	ranger: a class-only forest is rejected with advice	1	pass
	randomForest: the risk is the probability of class 1	1	pass
	a forest with more than two classes is rejected	1	pass
	bayespred: the risk is the posterior mean of risk	3	pass
	a numeric or logical vector is rejected	2	pass
	an object without a usable predict() gets a clear error	1	pass
	default method works for a model with predict(type = ‘response’)	1	pass
	missing covariates are caught	1	pass
test-prediction-routes.R	predict_function takes precedence over voipred’s own method and the standard route	3	pass
	predict_function gets horizon and cause; it is checked like any model	12	pass
	output shapes are read generically	10	pass
	standard binary route: type = ‘response’, then ‘prob’, then riskRegression	2	pass
	standard survival route: riskRegression::predictRisk, cause by label or position	2	pass
	a model with no usable route gets one clear error with the ways out	3	pass
	a new survival model works with predict_function	3	pass
	predict_risk() is internal	2	pass
test-probabilities.R	only 0s and 1s are rejected as classes, not probabilities	3	pass
	integer 0/1 predictions are rejected too	1	pass
	probabilities that include some exact 0s and 1s are accepted	2	pass
	a random forest regression on a 0/1 outcome still works	2	pass
	a glm asked for log-odds would be outside [0, 1]: predict_risk uses probabilities	2	pass
	values above 1 or below 0 are rejected as not probabilities	3	pass
	log-odds from a glm would be rejected as scores	1	pass
	a linear probability model (lm) with predictions below 0 is rejected	2	pass
test-risk-input.R	binary: risks given directly give the same results as the model	11	pass
	survival and competing risks, with Kaplan-Meier and cIPCW	4	pass
	a model’s risks are predicted once and reused by every later step	2	pass
	`predicted_risk` is checked	10	pass

File	Test	Checks	Result
test-survival-competing.R	riskRegression::CSC for both causes, with numeric or labelled causes (standard route)	21	pass
	riskRegression::FGR (standard route) and cmprsk::crr (voipred method) agree	18	pass
	randomForestSRC competing-risks forest (standard route, cause by position)	14	pass
	survival's multi-state coxph: Aalen-Johansen from cause-specific Cox models	17	pass
	multi-state coxph: causes matched by label or position, and the checks	5	pass
	a Fine-Gray model for another cause is rejected (riskRegression would not notice)	3	pass
	a standard survival model with a competing outcome works, with a warning	4	pass
	a competing-risks model needs a competing-risks outcome	10	pass
	missing covariates are reported for competing-risks models	4	pass
	numeric competing-risks codes get a hint to use a factor status	1	pass
test-survival-km.R	KM net benefit equals dcurves at every threshold, including NA thresholds	71	pass
	the risk by the horizon (treat all) is 1 - Kaplan-Meier	1	pass
	the internal KM and Aalen-Johansen match survival::survfit, with weights	9	pass
	no follow-up to the horizon among the treated gives NA and a warning	3	pass
	nobody treated gives NB_model = 0	1	pass
	a risk at the threshold is treated and a risk of 0 never is	2	pass
	weights: all ones equal no weights; integer weights equal duplicated rows	3	pass
	Aalen-Johansen net benefit equals dcurves for both causes	7	pass
	with competing risks, weights equal duplicated rows	1	pass
	KM is unbiased with independent censoring and biased with informative censoring	2	pass

File	Test	Checks	Result
test-survival-models.R	coxph fitted with x = TRUE (Efron, Breslow, case weights, ties, strata, splines)	30	pass
	rms::cph and rms::psm	12	pass
	survreg (Weibull, log-normal, log-logistic, exponential, strata)	32	pass
	flexsurvreg and flexsurvspline (formula written in the call or stored in a variable)	19	pass
	random survival forests: ranger and randomForestSRC	12	pass
	prodlim	6	pass
	coxph without x = TRUE (riskRegression refuses it): own baseline hazard	32	pass
	a coxph model that needs its (unavailable) data gives a clear error	1	pass
	the own Cox baseline hazard equals survival::basehaz	2	pass
	aorsf oblique survival forest	7	pass
	partykit::ctree survival tree	7	pass
	rstpm2::stpm2 flexible parametric model (needs rstpm2 attached)	7	pass
	mgcv::gam with the Cox PH family	9	pass
	models giving only relative risks or times have no route: clear error	6	pass
	a binary model with a survival outcome works, with a warning	5	pass
	survival models with a binary outcome are rejected	9	pass
	a horizon beyond the model’s own follow-up gives a warning	6	pass
	missing covariates are reported for survival models	11	pass
	each model’s risk grows with the horizon	2	pass
test-survival-outcome.R	the survival outcome can be given as a formula, a Surv object or a column	5	pass
	a binary outcome can also be given as a formula	1	pass
	horizon is required and must be a positive number within follow-up	6	pass
	survival-only arguments are rejected for a binary outcome	4	pass
	cause is only for competing risks and must be an event level	2	pass
	only right-censored outcomes are supported	2	pass
	missing or negative times and missing statuses are rejected	5	pass
	censoring settings must be consistent	4	pass
	cipcw_control() checks its inputs	11	pass
	cIPCW weights must have one value per row	1	pass
	print shows the outcome type, horizon and estimator	2	pass

File	Test	Checks	Result
test-voi-refit.R	every known censoring model is refitted with the draw’s weights	24	pass
	other censoring models need refit_function	5	pass
	refit_function replaces the automatic refit	3	pass
	models validated with the same censoring settings share one refit per draw	1	pass
	EVPI, P(useful) and EVSI with the refit, also for competing risks	6	pass
	given censoring weights work in nb() but not in the value of information	5	pass
	refit settings are checked	5	pass
test-voi.R	draws with all weights 1 reproduce nb() for every outcome and estimator	12	pass
	EVPI and P(useful) equal predtools::evpi_val() (Sadatsafavi et al. 2023)	17	pass
	EVSI equals the bootstrap algorithm of Sadatsafavi et al. (2025)	5	pass
	EVSI is between 0 and EVPI, grows with the future sample, and reaches EVPI	9	pass
	P(useful), P(best) and relative EVPI follow their definitions	14	pass
	relative EVPI is Inf when no model is better than the defaults but EVPI > 0	2	pass
	a survival outcome without censoring before the horizon gives the binary results	4	pass
	survival, competing risks and cIPCW: EVPI, EVSI, P(useful), P(best) run and are coherent	6	pass
	thresholds where the net benefit cannot be estimated give NA, with a warning	2	pass
	case weights are part of the draws	1	pass
	inputs are checked	15	pass